

Parvez Akhtar

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PROFESSIONAL SUMMARY

Senior Applied/GenAI Engineer with 5+ years building production-ready systems, specializing in AI agents that turn natural language into working software. Builds multi-step agent workflows with tool/function calling, context management (sub-agents, retrieval-based context, sliding windows), and structured outputs across multiple LLM providers (OpenAI GPT, Anthropic Claude). Data-driven on prompt engineering and automated LLM evaluation, having cut inference costs by ~90% while taking experimental AI to reliable enterprise production.

TECHNICAL SKILLS

GenAI & Agents: AI Agents & Multi-Agent Orchestration (LangGraph, OpenAI Agents SDK, Anthropic SDK, LangChain), Multi-Step Workflow Orchestration, Context Management (Sub-Agents, Retrieval-Based Context, Sliding Context Windows, Agent Memory), Tool/Function Calling, MCP (Model Context Protocol), Structured Outputs, Human-in-the-Loop, Guardrails & Prompt-Injection Defense, Prompt Engineering, RAG (Retrieval-Augmented Generation: Hybrid Search, Re-ranking), LlamaIndex, Fine-tuning (LoRA/QLoRA, Hugging Face), Embeddings, Vector DBs (Qdrant/PGVector).

LLM Providers: OpenAI (GPT series), Anthropic (Claude), Azure OpenAI, AWS Bedrock; quantitative model evaluation and selection per task.

Languages: TypeScript, Python, Java, SQL.

Frameworks: FastAPI, PyTorch, TensorFlow, Scikit-learn, XGBoost, Spring Boot, Django.

Cloud & Infra: AWS (SageMaker, Bedrock, Lambda, ECS), Azure OpenAI, Kubernetes, Docker, Terraform, Temporal.io.

Data Engineering: Snowflake Cortex, Apache Airflow, Kafka, Elasticsearch, ETL Pipelines, MongoDB, Redis, PostgreSQL, Browserbase.

LLMOps & Evaluation: LLM Evaluation Benchmarks & Tracing (Langfuse, RAGAS, OpenTelemetry), failure-mode analysis, SigNoz, ELK Stack, CloudWatch, Prometheus.

PROFESSIONAL EXPERIENCE

Bigstep Technologies

Remote

Senior AI/ML Engineer, Lead Architect

04/2025 – Present

- As **Lead Architect**, drove cloud-native GenAI delivery across client accounts spanning high-growth startups (Seed to Series F) and global enterprises including **CNBC London** and **VTS**, bringing clarity to ambiguous business problems and translating them into production architectures.
- **Client: VTS (current), AI Agent for Document Intelligence:** Built an **AI agent** that abstracts structured terms from commercial real-estate leases with **structured-output validation**. Diagnosed a **10-point accuracy regression** through systematic **failure-mode analysis** of multi-turn outputs, redesigning prompt strategy and image preprocessing to recover the full drop and restore field-level extraction to its **~88% baseline**, unblocking downstream automation.
- **Tool Use & Workflow Orchestration:** Orchestrated a serverless **multi-agent** framework for Salesforce automation with **tool/function calling** over **MCP (Model Context Protocol)** and **swarm-based, multi-step orchestration**, letting agents safely call external APIs and run database queries. Cut idle compute via a pure **pay-per-execution** model (saving **~\$45k/year**).
- **Client: Outcomelabs.ai (GenAI Platform):** Architected a **Field-Aware Extraction Engine** processing **2M+ records/month**. Built a **"Strategy Router"** that classifies each field by complexity and routes ~60% to deterministic rule-based parsers, reserving the LLM for the rest, quantitatively choosing the cheapest method per task and cutting token spend ~40% with no measurable accuracy loss.
- **Search & Ranking:** Built a **Hybrid Search** pipeline fusing **SPLADE** (learned sparse) and **Sentence-BERT** (dense) retrieval via **Reciprocal Rank Fusion**, then layered **Cross-Encoder re-ranking** over the top candidates, improving **NDCG@5** by **~15 points (0.71 to 0.86)**.
- **Scale & Impact:** Scaled the platform from MVP to production, underpinning a **\$20M+ enterprise deal pipeline** within 6 months while optimizing fine-tuning strategies for cost efficiency.
- **Core Engineering:** Built and curated datasets for an entity matching pipeline on 22M records using an ensemble of **TF-IDF/Cosine Similarity** and 7 text metrics feeding an **XGBoost classifier** (85.3% AUC).

Kuttl (Incubated at IIM Lucknow)

Remote

Founding Engineer

12/2023 – 03/2025

- Joined as **Founding Engineer** for an IIM Lucknow-incubated, GenAI-native orchestration platform; built the MVP from 0-to-1 that secured a term sheet. Documented and standardized the entire codebase and infrastructure, ensuring a **zero-downtime handover** of the platform to the incoming engineering team.
- Built the core backend and AI infrastructure from scratch, scaling the platform to handle peak loads while maintaining strict startup cost constraints.

- **Quantitatively evaluated and selected models per task:** fine-tuned **Llama-3-8B** (using LoRA) to replace GPT-4 for structured entity extraction, achieving **98% schema compliance** on **structured outputs** while reducing per-token inference costs by **90%** and latency by 300ms.
- Engineered a distributed document parsing pipeline using **FastAPI** and **Temporal.io**, processing 50k+ daily messages across multi-turn conversations. Enforced resilient retry logic and dead-letter queues to ensure no data loss.
- Operationalized the ML lifecycle with **automated LLM evaluation benchmarks** in **CI/CD gates (Ragas)** scoring **faithfulness and answer-relevancy**, automatically blocking deployments that fell below threshold and catching quality regressions before production.
- Applied semantic caching and request batching for RAG endpoints, improving **P99 latency by 40%** during high-concurrency bursts.

Amplicom

Bengaluru, India

SDE-2

05/2023 – 12/2023

- Led a team of 3 through a **Strangler-Fig migration** of a legacy monolith to Spring Boot microservices, enabling independent, incremental deploys with instant rollback; enforced **SonarQube** quality gates that cut cyclomatic complexity ~42%.
- Established a centralized identity broker using **Keycloak** and **Spring Security**, replacing stateful DB lookups with **stateless JWT** validation to cut login latency by 30%.
- Engineered a **Schema-per-Tenant** architecture with **RBAC** for 21 enterprise clients, ensuring data integrity via automated checksum reconciliation pipelines.

Amazon

Bengaluru, India

Software Development Engineer (SDE)

03/2021 – 03/2023

- Owned the full SDLC for high-scale microservices, co-developing the SmartPOS billing engine and scaling it to support **3,000+ enterprise merchants** with 99.99% availability.
- Optimized cloud infrastructure costs by **25% (~\$100k/yr)** via strategic re-architecture of ECS services to utilize **Spot Instances** and custom auto-scaling policies.
- Developed a data-driven testing framework and **CloudWatch** service-level alarms, identifying **N+1 query bottlenecks** that cut p95 API latency by **20%** and surfaced performance regressions before release.
- Executed the zero-downtime migration of 6 core microservices from GCP to AWS, ensuring data consistency and seamless customer transition.

Publicis Sapient

Remote, India

Associate Data Engineer

09/2020 – 03/2021

- Designed high-frequency data ingestion pipelines using **Apache Airflow** and **Crawlera** to scrape pricing data from Amazon, Walmart, and Target, bypassing complex anti-bot protections for competitive intelligence.
- Refined **Elasticsearch** indexing strategies and aggregation queries, reducing query latency by ~40% to enable sub-second dashboard load times for brand performance analytics.

EDUCATION

National Institute of Technology Delhi

Delhi, India

B.Tech in Computer Science & Engineering

2016 – 2020

ACHIEVEMENTS

Amazon "On the Hero Road" Award: Recognized for exceptional impact on team delivery and engineering excellence.

Udacity Computer Vision Nanodegree: Full scholarship recipient (sponsored by Facebook).

Deep Learning Specialization: Completed 5-course specialization (Coursera) with perfect score.